EDUFLOW: REGRESSION-BASED PREDICTIVE ENROLLMENT SYSTEM FOR COLEGIO DE STA. TERESA DE AVILA

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In Partial Fulfillment
of the Requirements for the Degree
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Bangcaya, Blake Justin C.
Castaneda, Tricia Mae B.
Dela Cruz, Gerah Misael
Estares, Alexander A.
Florano, John Cris C.
Gallardo, Kim Charles M.
Jacob, Amiel A.
Lim, Anjelo Miguel
Mondejar, Jeremy C.
Pascasio, Franz Simon S.
Perez, Menard
Samson, Joshua Nicus M.
Trias, James J.

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APPROVAL SHEET

In partial fulfillment of the requirements for the degree **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY,** this Capstone project entitled **"EDUFLOW: REGRESSION-BASED PREDICTIVE ENROLLMENT SYSTEM FOR COLEGIO DE STA. TERESA DE AVILA"**, has been prepared and submitted by **(Group Members),** who are hereby recommended for project presentation.

DR. HAROLD R. LUCERO
Capstone Project Mentor

Approved by the Committee for Project presentation with a notation of _____ on (date of defense).

XXXXXXXXXXXXXXX,
Chairperson, Project Presentation Committee

XXXXXXXX
Panel Member

YYYYYYY
Panel Member

Accepted and approved in partial fulfillment of the requirements for the degree **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**.

ISAGANI M. TANO, PHD-ELM, DIT
Dean of College of Computer Studies
Date: _____

DEDICATION

Proponents of this project first dedicate it to the Almighty God, the unwavering foundation, fount of motivation, sagacity, erudition, and discernment. The proponents recognize that You have presented us with this trial to facilitate not only our scholarly growth but also our personal development.

Secondly, the proponents dedicate this to our friends and classmates for their collaboration throughout the course of this project. We express our gratitude to Dr. Harold R. Lucero, who has been a guiding force, providing us with encouragement and direction from the outset.

Finally, we dedicate this to our family, the ones who have provided unwavering strength and encouragement to see this study through to completion, serving as our foremost supporters when it comes to education.

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EXECUTIVE SUMMARY

Title: **Eduflow: Regression-Based Predictive Enrollment System For Colegio De Sta. Teresa De Avila**

Proponents: **Bangcaya, Blake Justin C.**
Castaneda, Tricia Mae B.
Dela Cruz, Gerah Misael
Estares, Alexander A.
Florano, John Cris C.
Gallardo, Kim Charles M.
Jacob, Amiel A.
Lim, Anjelo Miguel
Mondejar, Jeremy C.
Pascasio, Franz Simon S.
Perez, Menard
Samson, Joshua Nicus M.
Trias, James J.

Instructor: **Dr. Harold R. Lucero**

Degree: **Bachelor of Science in Information Technology**

Date Completed: **December 2025**

As the 21st century requires digitization, it is high time to get the enrollment also digitized using automation in student appointment and course enlistment. Catalyst to address the inefficiencies in the conventional in-person registration process - Development of Quezon City University Enhanced Grade Slip Access and Registration Process. Based on the improved NBI application process, this system will be serving the process of enlistment for the existing and

continuing students of QCU. As a catalyst, the system aims to minimize steps into two (2), first is the Online Appointment by filling out the Student Application Form, and second, Identity confirmation through Reference Number, and Printing of Enrollment Certificate, and Go. This system breaks the problems of no specific number of enrollees to be anticipated in a day, requiring safety and order personnel for thousands of enrollees waiting in seats for their enrollment to be processed, and section allocation issues. The system project is exclusive within the university premises, focusing on appointment, scheduling, and evaluation of students and administrative staff as the main beneficiaries. The intuitive interface performs better than the current process increasing satisfaction between students and staff, it is also time-saving with the new 2-way process, and data security is ensured with a reduced labor force because of automation features that can be modified on a case-by-case basis. By leveraging technology to address these challenges, institutions like Quezon City University can not only enhance operational efficiency but also deliver a superior experience for both applicants and existing students.

Keywords: Appointment, Enrollment, Registrar, Registration, Student Enlistment Form

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CHAPTER 1

Capstone Project Background

This chapter introduces the foundation of the study by presenting the project's background, purpose, and objectives. It provides an overview of the problem being addressed and the significance of developing the proposed system. The chapter outlines both the main and specific objectives that guide the project's direction, along with its scope and limitations to define its coverage and constraints. It also discusses the theoretical and conceptual frameworks that serve as the basis for the study's development and implementation. Lastly, key terms used throughout the research are defined to ensure clarity and consistency of understanding.

Introduction

In the modern digital environment, predictive analytics and automated enrollment platforms play an increasingly vital role in improving how educational institutions operate (Maliberan, 2021; Masinading, Dalagan, & Manib, 2024). Using statistical and machine learning methods to forecast enrollment enables schools to plan faculty assignments, facilities, and other resources more efficiently—particularly when student numbers change from year to year. For private institutions such as Colegio de Sta. Teresa de Avila (CSTA), which offers programs including the Bachelor of Science in Information Technology (BSIT), Bachelor of Science in Hospitality Management (BSHM), Bachelor of Science in Tourism Management (BSTM), Bachelor of Secondary Education (BSED) in English, Filipino, Social Science, and Mathematics, and the Bachelor of Elementary Education (BEED) in General Education, accurate

enrollment forecasting is more than a convenience; it is essential for stable operations and informed decision-making.

In the Philippine context, previous research highlights the value of predictive modeling for enrollment planning. Maliberan (2021) employed regression analysis and multiplicative decomposition to project student numbers at Surigao del Sur State University, reaching a high accuracy rate (Mean Absolute Percentage Error ≈ 3.10%) based on historical records. Likewise, Masinading, Dalagan, and Manib (2024) applied time-series forecasting for semestral enrollment at Davao Oriental State University, identifying upward trends that supported resource allocation and budgeting. Villafuerte (2023) examined the "myMapua" online enrollment platform, uncovering issues in user satisfaction, interface design, and system reliability that hindered efficiency. Similarly, Rodriguez (2024) explored the development of an online enrollment and student-tracking system for Vinzons Pilot High School, demonstrating how digital solutions streamline processes once handled manually, reducing both delay and error.

At Colegio de Sta. Teresa de Avila, comparable challenges persist. Enrollment figures have fluctuated—approximately 4,500 in 2015, a sharp decline during the pandemic years, and a recovery to eight IT sections by 2025. The present workflow still depends on several manual or semi-manual stages such as interviews, document submission, assessment, payment, sectioning, ID capture, and medical clearance. These overlapping steps often cause inconsistencies, including mismatched unit costs across departments and scheduling conflicts for classrooms, leading to delays, billing mistakes, resource imbalance, and frustration for students and parents alike.

Although local research has validated the effectiveness of regression and decomposition models in predicting university enrollment (Maliberan, 2021; Masinading et al., 2024) and highlighted usability problems in existing systems (Villafuerte, 2023), few have merged these approaches with a modular, integrated workflow tailored for institutions like CSTA. No prior work has combined forecasting with assessment, billing, scheduling, and reporting modules suited to a private college setting in the Philippines.

For this reason, the researchers developed EDUFLOW, a regression-based predictive enrollment system designed for CSTA. The project aims to address both forecasting and administrative integration, enabling the institution to anticipate student numbers, lessen errors and processing time, optimize classroom and faculty usage, and enhance the enrollment experience overall. In academic terms, the study contributes to current literature by linking predictive analytics with a complete system framework for private higher-education institutions; in practical terms, it responds directly to CSTA's recurring enrollment fluctuations. With these issues and opportunities in mind, this study centers on meeting the forecasting and system-integration requirements of CSTA's enrollment process, serving as the foundation for the project context.

Project Context and Its Background

In the digital era, higher education institutions (HEIs) face increasing pressure to modernize their enrollment processes while ensuring efficient use of resources. Student enrollment, being the foundation of academic and financial planning, directly impacts

faculty allocation, classroom management, budgeting, and overall institutional sustainability. However, many schools in the Philippines continue to rely on manual or semi-manual enrollment procedures, which often result in delays, errors, resource mismatches, and poor user experiences for students and parents.

For Colegio de Sta. Teresa de Avila (CSTA), these challenges are particularly evident. Enrollment numbers have fluctuated significantly in recent years, with a dramatic decline during the pandemic and subsequent recovery in 2025. The current enrollment workflow involves numerous steps such as interviews, requirements submission, assessment, payments, sectioning, ID capture, and medical evaluation which, when handled manually, are prone to conflicts like billing discrepancies, room scheduling issues, and inefficient allocation of faculty. These inefficiencies pose risks not only to operational effectiveness but also to student satisfaction and institutional credibility

In response, the researchers propose EDUFLOW: A Regression-Based Predictive Enrollment System for CSTA. This system aims to combine statistical forecasting with modular administrative functions including assessment, billing and payments, room scheduling, and report generation within a unified platform. By doing so, EDUFLOW is expected to enhance enrollment accuracy, minimize delays and errors, optimize faculty and facility utilization, and provide a smoother experience for students and parents. Beyond addressing CSTA's immediate needs, this project contributes academically by bridging predictive analytics with holistic enrollment system design for private higher education institutions.

Purpose and Description

The primary purpose of this project is to develop a computer-based system that will assist the school in managing its enrollment processes and student records more efficiently. The system is intended to make enrollment procedures easier, faster, and

more organized for both new and returning students—from the submission of requirements to payment processing and report generation. It also aims to minimize manual work and reduce errors commonly encountered in paper-based transactions. The proposed system will handle essential school information such as student profiles, faculty lists, class schedules, payments, and reports. It will cater to the three academic programs offered by the institution: Bachelor of Science in Information Technology (BSIT), Bachelor of Science in Hospitality Management (BSHM), and Bachelor of Secondary Education (BSED).

Through this system, students can be accurately assessed, properly sectioned, and provided with complete and reliable enrollment forms. For new students, the process includes interviews, submission of requirements, assessment, payment, release of registration forms and blue cards, sectioning, ID capturing, and medical examinations. For continuing students, the process involves evaluation, assessment, payment, release of registration forms and blue cards, sectioning, ID validation, and medical check-ups. Furthermore, the system integrates an analytics feature that predicts the number of students expected to enroll in future terms. This predictive capability enables the school to plan ahead for classroom availability, teacher assignments, and resource allocation. In addition, the system allows staff to easily update and manage institutional information, such as rooms, programs, departments, and facilities, ensuring that data remains accurate and up to date.

Overall, this project seeks to provide a dependable and user-friendly school management system that meets the needs of both students and staff. Automating key administrative processes enhances efficiency, reduces workload, and supports a smooth and well-organized enrollment experience.

Objective of the Study

The general objective of this study is to design and develop a regression-based predictive enrollment system, "EDUFLOW," for Colegio de Sta. Teresa de Avila (CSTA). The system aims to streamline enrollment operations, predict student population trends, and effectively handle institutional, faculty, and student data.

Specifically, it aims to:

1. Design and develop the EDUFLOW system capable of:
   a. integrating major modules such as Student, Faculty, Curriculum, Schedule, Payment & Billing, Assessment, Report Generation, Analytics, and File Maintenance to manage and organize enrollment-related data efficiently;
   b. automating the enrollment workflow for both new and returning students to minimize manual errors, delays, and inconsistencies; and
   c. providing a user-friendly interface that supports accurate, fast, and secure transactions.
2. Integrate a regression-based predictive analytics feature within EDUFLOW to analyze historical enrollment data, identify trends, and forecast future student numbers to assist CSTA in making data-driven academic and administrative decisions.
3. Implement a Payment & Billing module that allows multiple payment methods such as GCash, Cash, Credit Card, and Bank Deposit, with a required minimum initial payment of ₱3,000, while excluding cheque transactions for security and convenience.

4. Enhance the data management and reporting features by ensuring that institutional information—such as fees, programs, departments, and faculty records—is well maintained, and that automated reports like registration forms, class lists, and enrollment summaries can be easily generated.
5. Evaluate the developed system's quality and user acceptance by:
   a. applying the Technology Acceptance Model (TAM) to measure perceived usefulness, ease of use, attitude toward use, and behavioral intention; and
   b. assessing the system's compliance with ISO/IEC 25010 software quality standards, focusing on functional suitability, performance efficiency, compatibility, usability, reliability, security, and maintainability.

Scope and Limitation/Delimitations of the Project

The EduFlow Enrollment System is developed to help Colegio de Sta. Teresa de Avila (CSTA) improve and streamline its enrollment process. This system will automate crucial tasks such as student registration, assessment, class sectioning, and payment handling. Additionally, it will incorporate data analytics using regression models to analyze past enrollment patterns and forecast future trends. Through this, the system aims to assist school administrators in planning better by identifying peak registration times, estimating enrollment numbers, and suggesting ways to manage resources like classrooms and faculty.

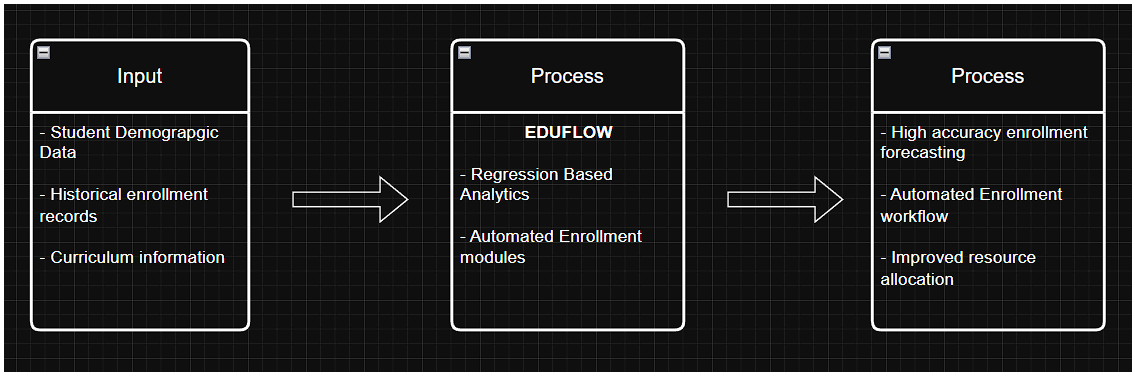
However, this study is focused strictly on the enrollment-related activities and does not cover functionalities beyond that. For example:

- The system will not cover academic grading or performance tracking, as its main function is to facilitate enrollment steps—from submitting requirements to assigning classes.

- It does not include features for online learning, attendance monitoring, or managing class activities.
- Accepted payment options are limited to GCash, cash, credit cards, and bank deposits; cheque payments are excluded, and bank deposits require manual confirmation.
- While the system supports assigning classrooms and schedules, it will not manage room maintenance or cleanliness.
- This is a web-based platform only, with no mobile app or connection to external student portals.

These boundaries help focus the project on enhancing the enrollment process, making it more efficient and accurate, while providing valuable analytical insights tailored to CSTA's current needs.

Theoretical Framework and Conceptual Framework of the Study



The EDUFLOW system is built on the combined ideas of Information Systems Design and Predictive Analytics Application, as reflected in the studies of Maliberan (2021), Masinading, Dalagan, & Manib (2024), Valencerina, Dela Vega, & Santos (2020), and Chamilco et al. (2023). These works show that when technology-based frameworks are enhanced with forecasting tools like regression models, schools can manage operations more effectively through accurate and timely insights on enrollment trends. The concept behind EDUFLOW is that by merging system efficiency with data forecasting, traditional manual processes can evolve into predictive, adaptive, and automated systems.

Previous studies also support this idea. Basilan & Baesa (2019) and Valencerina et al. (2020) found that ICT-based enrollment systems improve efficiency by minimizing human error and reducing processing time. Maliberan (2021) and Masinading et al. (2024) demonstrated how regression and time-series models can make enrollment predictions more precise. Rodriguez (2024) and Villafuerte (2023) focused on ensuring that student enrollment systems are user-friendly and reliable, while Chamilco et al. (2023) highlighted the importance of descriptive analytics in decision-making. Meanwhile, Soriano (2023) and Hayagan (2022) emphasized the role of proper data management and transparency in school administration. Altogether, these studies show that integrating predictive modeling, analytics, and automation leads to smarter decisions and a smoother user experience.

Based on these insights, the conceptual framework of EDUFLOW is designed around three main components: (1) System Automation, (2) Predictive Analytics, and (3) Administrative Decision Support. Automation manages essential tasks such as student registration, billing, and assessment (Basilan & Baesa, 2019; Valencerina et al., 2020). Predictive Analytics uses regression models to forecast enrollment trends (Maliberan, 2021; Masinading et al., 2024). Lastly, Decision Support helps administrators plan and make data-driven reports (Chamilco et al., 2023; Soriano, 2023). Combined, these components create an integrated system that enhances operational efficiency, ensures data accuracy, and empowers Colegio de Sta. Teresa de Avila to anticipate enrollment changes and manage resources effectively.

Definition of Terms

Predictive Analytics. Maliberan (2021) explains that predictive analytics refers to statistical or machine learning techniques used to forecast future events, such as student enrollment numbers, based on historical data.

Automated Enrollment System. Rodriguez (2024) defines an automated enrollment system as a technology platform that digitizes and streamlines manual processes involved in student registration and tracking within educational institutions.

Regression Analysis. Maliberan (2021) describes regression analysis as a statistical method used to examine relationships among variables and project values, widely applied in predicting student enrollment trends.

Multiplicative Decomposition. Maliberan (2021) states that multiplicative decomposition is a technique in time-series analysis separating trend, seasonal, and irregular patterns to enhance forecasting accuracy.

Time-Series Forecasting. Masinading, Dalagan, and Manib (2024) discuss time-series forecasting as the use of past data points to predict future values, aiding educational institutions in resource planning.

Online Enrollment Platform. Villafuerte (2023) relates that an online enrollment platform provides students and administrators with a digital interface for managing admissions, registration, assessment, and payment processes.

Usability. Villafuerte (2023) explains that usability is the measure of how efficiently users can learn, navigate, and complete tasks within a system or software.

Workflow Integration. Rodriguez (2024) refers to workflow integration as the seamless connection of multiple administrative processes, such as assessment, sectioning, and billing, within a unified digital system.

Room Scheduling. Rodriguez (2024) defines room scheduling as the process of allocating physical spaces for classes and events, often managed through specialized software to avoid conflicts.

System Modularization. Masinading et al. (2024) explains that system modularization is the architectural design wherein different functions (modules) in an information system operate independently but integrate for overall efficiency.

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